Project Title	**Zomato Order & Restaurant Analysis Using Power BI**
Skills take away From This Project	• **SQL Data Import & Transformation** • **Data Cleaning & Preprocessing** • **Data Modeling in Power BI** • **DAX Functions & Calculations** • **Interactive Visualizations** • **Business Insights & KPI Analysis**
Domain	**Food & Beverage, Business Analytics**

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Problem Statement:

Analyze and visualize Zomato's restaurant and order data to uncover trends in customer preferences, restaurant performance, pricing impact, and location-based insights. The goal is to transform raw data into actionable business intelligence for decision-making.

Business Use Cases:

- Identifying top-performing restaurants based on ratings and order volume.
- Analyzing how pricing impacts customer orders.
- Understanding customer preferences in different cities and areas.
- Optimizing delivery times and improving efficiency.
- Forecasting demand trends for future growth.

Approach:

Task 1: Upload Dataset into MySQL

1. Install MySQL and create a new database (`ZomatoDB`).
2. Create two tables:
    - `Zomato_Restaurants`
    - `Zomato_Orders`
3. Import the **Zomato_Orders.csv** and **Zomato_Restaurants.csv** files into MySQL.

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Task 2: Perform Simple Data Transformations Using SQL

Learners should perform the following transformations using SQL queries:

1. **Basic Data Cleaning**
    - Remove duplicate records.
    - Handle NULL values (if any) by replacing them with appropriate values.
2. **Data Exploration Queries**
    - Count the number of restaurants in each city.
    - Find the top 5 cities with the highest number of orders.
    - Calculate the total revenue generated by each restaurant.
3. **Data Aggregation**

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- Find the average order amount for each city.
- Identify the top 5 restaurants with the highest total sales.

4. **Data Joins**
   - Join the `Zomato_Orders` and `Zomato_Restaurants` tables to get restaurant names along with order details.
5. **Export the transformed dataset**
   - Save the transformed data as an Excel file.

Task 3: Power BI Analysis and Visualization

1. **Import Data into Power BI**
   - Load the Excel file containing transformed data.
2. **Basic Data Cleaning in Power Query Editor**
   - Ensure the data types are correctly set.
   - Remove unwanted columns if needed.
3. **Modeling Data Relationships**
   - Establish a relationship between `Restaurant_ID` in both datasets.

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Power BI tasks

🔹 Task 1: Number of Restaurants per City

Objective:

Analyze the distribution of restaurants across different cities to identify which cities have the highest and lowest restaurant presence.

Task Explanation:

- Extract the total count of restaurants for each city from the dataset.
- Use a **bar chart** or **map visualization** to display the distribution.
- Identify trends such as **high restaurant density areas** and **underrepresented locations**.
- Analyze whether restaurant availability correlates with city population or food demand.

Expected Output:

A bar chart or map showing the **number of restaurants per city** with clear insights on restaurant density.

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🔹 Task 2: Percentage of Orders from Different Cities

Objective:

Understand which cities contribute the most to total orders and how demand varies across regions.

Task Explanation:

- Aggregate the total number of orders for each city.
- Create a **pie chart** to represent the percentage of total orders by city.
- Compare the order distribution against restaurant density to check for imbalances.
- Provide insights into which cities dominate the Zomato order market.

Expected Output:

A **pie chart** that visually breaks down **order distribution by city** for easy comparison.

🔹 Task 3: Order Amount Trends Over Time

Objective:

Analyze how order amounts fluctuate over time to identify peak sales periods and slow periods.

Task Explanation:

- Extract date-wise order amounts from the dataset.
- Use a **line chart** to visualize total sales over time.
- Identify **trends, spikes, and seasonal variations** (e.g., festive seasons, weekends).
- Provide insights into when the highest and lowest order volumes occur.

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Expected Output:

A **line chart** showing the **order amount trends over time** with insights on peak sales periods.

🔹 Task 4: Correlation of Factors Affecting Average Rating

Objective:

Identify correlations between different factors (e.g., **price range, total ratings, delivery time**) and **average restaurant rating** to understand customer preferences.

Task Explanation:

- Select key numerical factors (**price, total ratings, delivery time, order count, etc.**) that might influence ratings.
- Use **scatter plots, correlation matrices, or trend lines** to visualize relationships.
- Identify any **positive or negative trends** (e.g., Does higher pricing lead to better ratings? Do shorter delivery times result in higher satisfaction?).
- Provide **insights from the visualization** regarding which factors have a significant impact on ratings.

Expected Output:

A **scatter plot or correlation matrix** showing relationships between factors like **price, delivery time, total ratings**, and **average rating**.

🔹 Task 5: Top 5 Restaurants by Total Sales

Objective:

Identify the **highest-earning restaurants** based on total revenue.

Task Explanation:

- Sum the total sales amount for each restaurant.
- Identify the **top 5 revenue-generating restaurants**.
- Use a **column chart** to compare their sales performance.
- Provide insights into what makes these restaurants perform better than others (e.g., higher pricing, more orders, premium reputation).

Expected Output:

A **column chart** displaying the **top 5 highest-earning restaurants**.

🔹 Task 6: Revenue by Area Using a Tree Map

Objective:

Visualize the revenue contribution from different areas to understand high and low-performing localities.

Task Explanation:

- Aggregate the **total revenue generated** by each area (locality).
- Use a **tree map** to represent revenue contribution visually.
- Identify **high-revenue and low-revenue areas**.
- Analyze if revenue is driven by a high number of orders or high average order value.

Expected Output:

A **tree map** showing the **distribution of revenue across different areas** with insights into top-performing localities.

🔹 Task 7: Order Density by City Using a Heat Map

Objective:

Analyze which cities have the highest concentration of orders.

Task Explanation:

- Group the number of orders by city.
- Use a **heat map** to visually represent order density.
- Identify **cities with the highest and lowest order volumes**.
- Compare order density with the number of restaurants in each city to see if supply meets demand.

Expected Output:

A **heat map** visualizing **which cities receive the most and least orders**.

🔹 Task 8: KPI Cards – Total Revenue & Average Order Value

Objective:

Display key business performance metrics such as **total revenue and average order value**.

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Task Explanation:

- Calculate **total revenue** and **average order value**.
- Create **KPI cards** to highlight these values.
- Provide a comparison (e.g., % increase or decrease from the previous period).
- Ensure that KPIs are **easy to interpret and visually appealing**.

Expected Output:

KPI cards displaying **total revenue** and **average order value**, possibly with trend indicators.

🔹 Task 9: Restaurant-Wise Sales Summary in a Table

Objective:

Create a structured **table summarizing restaurant-wise sales performance**.

Task Explanation:

- Display restaurant names, total sales, number of orders, and average order value.
- Apply **sorting and conditional formatting** to highlight best-performing and worst-performing restaurants.
- Allow filtering by city, area, or price range to drill down into specific restaurant categories.

Expected Output:

A **data table** summarizing **restaurant-wise sales performance** with useful sorting and filtering options.

🔹 Task 10: Power BI Dashboard – Combining All Visuals

Objective:

Create a **fully interactive dashboard** combining all previous visualizations for a complete business overview.

Task Explanation:

- Integrate **bar charts, pie charts, scatter plots, KPI cards, and heat maps** into one cohesive dashboard.
- Ensure **filters and slicers** allow users to drill down by city, area, or restaurant.
- Arrange visuals in a structured way for **clear insights and storytelling**.
- Optimize dashboard layout for a user-friendly experience.

Expected Output:

A **Power BI dashboard** combining all key metrics and visualizations into a **single interactive report**.

Results:

- A well-structured dashboard showcasing key performance metrics.
- Identification of high-revenue restaurants and customer ordering patterns.
- Location-based analysis of demand and restaurant density.
- Improved understanding of delivery time efficiency.

Project Evaluation metrics:

- Accuracy of SQL transformations and queries.
- Correct data modeling and relationship building in Power BI.
- Number and quality of visualizations created.
- Depth of insights extracted from data analysis.
- Presentation and storytelling effectiveness.

Technical Tags:

SQL,

Power BI,

Data Cleaning,

Data Modeling,

Visualization,

DAX,

Business Intelligence

Data Set:

The dataset consists of two files:

Zomato_Orders -updated

Zomato_Restaurants -updated

1. **Zomato_Orders.csv** – Contains information about customer orders, restaurant IDs, order values, delivery times, and ratings.
2. **Zomato_Restaurants.csv** – Contains restaurant details like name, location, cuisine type, and pricing.

Data Set Explanation:

Zomato_Orders.csv

Column Name	Explanation
order_id	Unique identifier for each order
restaurant_id	ID of the restaurant associated with the order
customer_id	Unique identifier for each customer
order_date	Date when the order was placed
order_time	Time when the order was placed
delivery_time	Time taken to deliver the order (in minutes)
total_cost	Total price of the order, including tax & delivery charges
item_count	Number of items in the order
payment_method	Payment mode used (Credit Card, UPI, COD, etc.)
customer_rating	Rating given by the customer for that order

Zomato_Restaurants.csv

Column Name	Explanation
restaurant_id	Unique identifier for each restaurant
restaurant_name	Name of the restaurant
city	City where the restaurant is located
area	Specific area in the city where the restaurant operates

cuisine	Type of cuisine offered (Indian, Chinese, Italian, etc.)
avg_rating	Average rating based on customer feedback
total_ratings	Total number of ratings received
price_range	Price range category (Low, Medium, High)
delivery_available	Whether delivery service is available (Yes/No)

Project Deliverables:

- **MySQL Database Export:** Transformed dataset after SQL processing.
- **Power BI Report:** A Power BI file containing visualizations and insights.
- **Project Documentation:** Explanation of findings and business recommendations.

Project Guidelines:

- Ensure clean data relationships between tables in Power BI.
- Use appropriate visualization techniques for each KPI.
- Avoid unnecessary duplicate data in reports.
- Provide well-structured dashboards with clear insights.
- Follow best practices in SQL query writing and Power BI modeling.

PPT Submission Instructions (Mandatory)

- Slide count must be **minimum 10 and maximum 15**
- No long paragraphs — **only bullet points**
- Include charts, icons, and images wherever useful
- Each task should be clearly covered in **one dedicated slide**

Timeline:

The project must be completed and submitted **within the Deadline.**

References:

Project Orientation Video	new-zomato-orietation.mp4
Capstone Explanation Guideline	Capstone Explanation Guideline
Project Live Evaluation	Project Live Evaluation
How to Submit the project	Project submission in zen p...-Recording…

PROJECT DOUBT CLARIFICATION SESSION (PROJECT, CLASS and CODE_KATA DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

BADM/BMAI & CAD

Timing: Tuesday, Thursday and Saturday (2:00PM to 3:00PM) [Except Second Saturday]

Session Link : BMAI/BADM_Project Doubt Clarification Session Link

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timings:

Weekday [Batches whose main-boot sessions happen on weekday] :

Monday-Thursday [No PLE on Friday]
Morning - 11AM to 12PM
Evening - 5PM to 6PM

Weekend [Batches whose main-boot sessions happen on weekends] :

1. Monday-Thursday [No PLE on Friday]
Morning - 11AM to 12PM
Evening - 5PM to 6PM
2. Saturday
Morning - 11AM to 12PM

Booking link : https://forms.gle/1m2Gsro41fLtZurRA [Old]
Booking link : http://bit.ly/4nXquBQ [New]

Approval Workflow

Created By:	Verified By:	Approved By:

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